

		$\approx 1.7 \times 10^{-3} \text{ m s}^{-1}$
17	D	Since $R = \frac{\rho L}{A}$, $R \propto \frac{L}{r^2}$, need $\frac{R}{2}$ implies $\frac{R}{2} = k \times 0.5 \frac{l}{r^2}$ (where k is a constant), Option A: $\frac{l}{\left(\frac{r}{2}\right)^2} = 4 \frac{l}{r^2}$ (incorrect) Option B: $\frac{l}{(2r)^2} = 0.25 \frac{l}{r^2}$ (incorrect) Option C: $\frac{2l}{\left(\frac{r}{2}\right)^2} = 8 \frac{l}{r^2}$ (incorrect) Option D: $\frac{2l}{(2r)^2} = 0.5 \frac{l}{r^2}$ (correct)
18	B	Q. No